

FROM THREAT SCORES TO DYNAMIC INSTITUTIONAL PRIORITIES

THE AUTONOMOUS CYBERSECURITY OPERATIONS CEN- TER: OPTIMIZATION WITH JUDGMENT

A Synthetic SOC Simulator and Its Transfer to Banking Regulation, Systemic Risk and Financial Stability

A pedagogical study of how telemetry, perception, threat assessment, contextual judgment, governance, bounded execution and feedback combine when environmental change forces priorities to override an original sequence—and why the same architecture applies when the “attack” is speculative rather than cyber.

CENTRAL PROPOSITION

Optimization chooses efficiently within a specified objective, feasible set and state of the world. Judgment becomes necessary when the environment changes and the institution must decide whether the original priority ordering still deserves to govern action. In the SOC, a high threat score does not automatically imply isolation: criticality, confidence, propagation, operational continuity, reversibility and delegated authority can change the sequence. In financial regulation, the analogous shock may be a run, speculative attack, liquidity spiral or market dislocation. The transferable architecture is therefore *optimization with judgment*.

Alejandro Reynoso

October 2026

AI for Financial Practitioners / Autonomous Systems / Institutional Design

Pedagogical Research Monograph

Educational simulator. The network, incidents, scores, actions and outcomes are synthetic teaching constructs. The financial transfer sections are architectural analogies and research hypotheses, not operational supervisory guidance.

Abstract

This monograph studies a synthetic autonomous Security Operations Center (SOC) as a laboratory for a broader class of self-driving institutions. The notebook models 100 endpoints, 20 servers, interdependent services, noisy telemetry and a sequence of synthetic incidents. The system does not reduce decision making to “detect and block.” It converts telemetry into evidence, estimates threat and propagation, prices the operational consequences of defensive actions, ranks feasible alternatives, subjects recommendations to a governance gate, executes only bounded actions and then observes the consequences. The central pedagogical result is that a threat score is evidence, not an instruction.

The architecture is especially useful because priorities can take precedence over a previously implied sequence. A technically aggressive response may rank highly on immediate containment yet be institutionally unacceptable when it interrupts identity, payments or another critical service. Conversely, waiting for certainty may increase propagation risk. Judgment therefore operates at the margin where multiple legitimate objectives conflict. Governance remains distinct: the system can recommend an action that it is not authorized to execute.

The second half of the monograph develops a transfer-learning exercise from cyber operations to finance. The transfer is conceptual and architectural rather than a claim that cyber-trained model weights should be reused. A bank run, speculative attack on a currency, liquidity spiral, margin shock or interconnected-institution failure can be represented as a changing network state with noisy signals, propagation channels, critical nodes, intervention costs, reversibility constraints and delegated authority. The regulatory problem then becomes one of dynamic priority ordering: preserve market discipline, protect liquidity, maintain payments, contain contagion, conserve public resources and avoid moral hazard—but not necessarily in the same order at every moment. The SOC notebook therefore provides a concrete model of *optimization with judgment*: algorithms preserve measurement and feasibility; judgment reorders priorities when context changes; governance constrains authority; and feedback continually reconstructs the state.

NOTEBOOK ARCHITECTURE

Telemetry → Perception → Threat assessment → Judgment → Governance gate → Defensive execution → Feedback → Reassessment. The notebook’s contribution is not any single score. It is the separation of these functions so that perception, recommendation, permission and execution remain auditable. **Executable notebook:** [Open the Google Colab notebook.](#)

Illustration omitted in this text-only edition: The notebook in one view: a synthetic institution, dynamic incident, judgment engine, governance gate, bounded execution and feedback.

1. Why a Cybersecurity SOC Is a Model of Institutional Judgment

A conventional optimization problem assumes that the objective is known. If the system knows the costs, constraints and state, it can select an action. Real institutions face a harder problem: the objective itself is often conditional on context. A SOC may prefer aggressive containment in ordinary systems but continuity in a payments server; it may tolerate uncertainty when actions are reversible but escalate when consequences are irreversible. The operating problem is therefore not only *which action minimizes a loss function?* It is also *which losses deserve priority now?*

This distinction connects the notebook to bounded rationality and organizational decision making. Simon's account of administrative behavior emphasizes that institutions act under limited information, multiple objectives and structured authority rather than through frictionless global maximization [15; 10]. Modern control theory similarly emphasizes repeated state estimation and feedback rather than one-shot plans [3; 14]. The notebook combines these intuitions with explicit machine governance.

THE CENTRAL DILEMMA

A suspicious payments server creates two simultaneous risks. Isolate too quickly and the defense can become the operational incident. Wait too long and the threat may propagate. The correct response depends on severity, confidence, systemic importance, propagation, operational cost, reversibility and authority. No single threat score contains all of that information.

The notebook is deliberately synthetic and defensive. Its intellectual value lies in making the decision architecture observable. There are no real targets or attack procedures. What matters is how the institution reacts when evidence changes.

The SOC is a useful institutional model because it must coordinate specialized functions that see different parts of the same event. Telemetry collection may detect an anomaly, an analyst may understand a service dependency, an optimizer may compare interventions, and an authorized manager may decide whether disruption is acceptable. Treating these as separate roles makes it possible to ask where information is lost, where assumptions enter, and who is accountable for a decision. A single end-to-end score hides those distinctions. It can make a decision appear objective even when the weights, data coverage or action permissions embody choices that should be visible to the institution.

The distinction between optimization and judgment can be stated without implying that either is superior. Optimization is valuable when objectives and constraints are sufficiently stable to compare alternatives consistently. It can rank actions, identify dominated choices and show how a recommendation changes when assumptions change. Judgment is needed when the institution must decide whether those assumptions still represent the situation. A priority order that was sensible under ordinary conditions may be inappropriate when a critical service is at risk or evidence quality deteriorates. Judgment does not mean discarding measurement; it means interpreting measurement against purpose, context and responsibility.

This framing also explains why the monograph uses the phrase “autonomous” cautiously. Autonomy here means that the system can carry out a structured cycle of observation, assessment and bounded action. It does not mean that the system possesses independent legal authority, understands institutional values in a human sense, or should be allowed to redefine its own mission. A system can autonomously monitor or recommend while requiring a person to authorize actions with large or difficult-to-reverse consequences. That graded autonomy is more realistic and more useful than a binary classification of “human” versus “machine.”

The central teaching question is consequently procedural as well as technical: can a reader trace a recommendation from evidence through assumptions, objectives and permissions to an observed outcome? If that chain is inspectable, a disagreement can be localized. Reviewers can challenge the signal interpretation, question the estimated operational cost, revise the priority ordering, or determine that the proposed action exceeded delegated authority. If the chain is hidden, a good outcome may be luck and a bad outcome may be impossible to diagnose. The notebook turns this traceability requirement into a small simulation that can be inspected and modified.

2. Cells 2–5: Build the Institution Before Asking It to Decide

Cells 2–5 establish reproducibility, assets, dependencies and telemetry state. Cell 2 fixes a random seed, defines 100 endpoints, 20 servers and a 120-tick horizon, and makes the optional language-model supervisor nonessential. This is more than software hygiene: reproducibility is a governance property because reviewers must be able to reconstruct the state that produced a decision.

Cell 3 defines assets by type, criticality, exposure, health, isolation status and service. The server population includes identity, payments, customer data, communications, analytics, operations, treasury and general services. Cell 4 then turns these assets into a dependency graph. Endpoints depend on services; servers also depend on one another. Criticality is therefore not merely an intrinsic property of a node. It interacts with network position.

Cell 5 introduces signals and incidents. The important conceptual move is that raw telemetry is not yet a decision. A signal has time, asset, category, anomaly and confidence. The institution must still aggregate and interpret it.

These setup choices define the experiment's world and its limits. A fixed seed makes two runs comparable when the policies being tested differ, while the synthetic asset population keeps the example small enough to inspect. The number of endpoints and servers is not intended as a calibrated representation of an actual enterprise. Instead, the combination creates enough variety to expose a recurring decision problem: many low-consequence observations coexist with a small number of services whose interruption would affect other parts of the institution. The simulation can therefore make operational consequences visible without pretending to reproduce a real network.

Reproducibility needs more than fixing a random seed. A reviewer should also know which parameters were used, which scenario was active, what state existed before an action and whether an optional component influenced the result. Those details distinguish a repeatable experiment from a merely repeatable program. The notebook's separation of initialization, scenario generation and decision functions helps make that distinction. An experiment can reset the state, rerun the same shock and then change one policy assumption at a time. Such discipline is useful in finance as well: a supervisory exercise should preserve the shock path and initial balance-sheet network when comparing alternative responses.

The asset table and dependency graph should be interpreted together. A high-criticality node may be important because of its own function, because many services rely on it, or both. A less prominent endpoint may become relevant if it lies upstream of a critical service. Conversely, a severe anomaly on a relatively isolated asset may have limited institutional reach. The graph is thus a compact representation of conditional importance rather than a simple ranking of machines. When transferring the idea, financial assets and institutions likewise need relationships: depositors, counterparties, collateral, payment obligations and common exposures can make local distress consequential beyond the affected entity.

Telemetry design determines what the institution is able to notice. Recording time permits sequences and acceleration to be examined; recording confidence prevents uncertain signals from being treated as facts; retaining the asset and category supports aggregation without erasing provenance. These fields do not solve perception, but they make it possible to reason about what the evidence represents. A practical lesson for the reader is to resist building a score before deciding what observations mean, how missing data will be represented and how evidence will be challenged. Otherwise, false precision enters at the foundation and is carried forward by every later stage.

Illustration omitted in this text-only edition: Cells 2–5 establish the synthetic institution, asset criticality, dependencies and the state needed to convert raw telemetry into an auditable decision problem.

TRANSFER PRINCIPLE 1: MODEL THE INSTITUTION, NOT ONLY THE SHOCK

A financial-stability system cannot reason from a price move alone. It needs institutions, balance-sheet criticality, funding dependencies, collateral links, payment-system roles and market connections. The SOC graph is a pedagogical analogue of that institutional map.

3. Cells 6–10: Actions, Scenarios, Perception and Threat Assessment

Cell 6 defines a bounded defensive action vocabulary: observe, increase monitoring, rate limit, isolate an endpoint, isolate a server or suspend a service. Each action carries intrusiveness, reversibility and an authority level. This is crucial. The action space is constructed before judgment is invoked, so a supervisor cannot invent arbitrary interventions.

Cell 7 defines a scenario generator with benign spikes, credential anomalies, coordinated incidents and critical-service anomalies. Cell 8 creates background telemetry and injects scenario events. Cell 9 performs perception by aggregating signals into evidence, while Cell 10 adds local propagation pressure and produces a threat assessment.

The sequence is pedagogically important:

$$\text{raw signal} \neq \text{evidence} \neq \text{threat} \neq \text{action}.$$

A mature autonomous institution preserves these distinctions. Collapsing them encourages automation bias: a model output is mistaken for a command.

The bounded action vocabulary serves two purposes. First, it makes alternatives comparable: observing, monitoring, rate limiting and isolation differ in their expected benefit, operational disruption and reversibility. Second, it makes authorization legible. If each action is assigned an impact and authority level in advance, the system can identify when a recommendation must be escalated instead of quietly expanding what it is allowed to do. A carefully designed action set should include a meaningful low-impact option, because a decision process that offers only “do nothing” and “disrupt service” cannot express proportionate responses.

Scenario generation is not simply a way to make a demonstration look realistic. It defines the cases against which the decision policy is being examined. A benign spike tests whether the process can avoid overreacting; a credential anomaly tests whether weak evidence can accumulate; a coordinated incident tests propagation; and an anomaly involving a critical service tests the conflict between containment and continuity. No finite set of scenarios proves that a policy will work in the field. It does, however, let the author state what kinds of situations the policy was designed to handle and reveal blind spots in that design.

Perception should preserve the relationship between observations and inferred conditions. Several weak signals may jointly support a stronger hypothesis, but aggregation can also conceal that all signals came from the same underlying source or depend on the same data pipeline. Confidence is therefore not an ornamental number: it indicates how much weight the institution should place on the inferred state and whether another measurement or human review is worth the delay. The threat assessment adds a different question by estimating potential consequence and propagation. Evidence can be highly reliable yet describe a low-impact event; uncertain evidence can still warrant attention if the plausible consequences are severe.

Separating the stages also creates meaningful checkpoints for evaluation. Perception can be tested for whether it identifies the intended evidence; assessment can be tested for whether it distinguishes severity and spread; and action selection can be tested for whether its response is proportionate. If the final decision is wrong, these checkpoints help determine whether the failure came from noisy inputs, a poor inference, an incomplete objective or an authority rule. A single end-to-end accuracy number would not reveal that diagnosis. The notebook’s staged structure is thus an educational design choice that encourages explicit reasoning about uncertainty rather than a claim that every stage can be measured perfectly.

Illustration omitted in this text-only edition: Cells 6–10 define bounded actions and dynamic scenarios, then transform noisy telemetry into evidence, propagation context and a threat estimate.

Why priorities can change the sequence

Suppose the system originally intends to monitor several medium-risk assets before acting. A new signal on a highly connected payments server changes the environment. The sequence should not remain sacred merely because it was previously rational. The new state can elevate containment, evidence gathering or escalation ahead of the original plan. The key is that the reordering is justified by institutional priorities, not by arbitrary model discretion.

4. Cells 11–15: From Optimization to Judgment and Governance

Cell 11 estimates operational cost. Intrusiveness is combined with asset criticality, dependency load and service importance. Cell 12 then ranks defensive alternatives using security benefit, operational cost, reversibility and an uncertainty penalty. This is the notebook’s explicit optimization-with-judgment layer.

A stylized action score can be written as

$$S(a) = \alpha B(a) - \beta C_{op}(a) + \gamma R(a) - \delta U(a),$$

where B is defensive benefit, C_{op} operational cost, R reversibility and U the uncertainty penalty. The notebook uses fixed pedagogical weights, but the deeper point is architectural: a technically effective action can lose once continuity and uncertainty are priced.

Cell 13 inserts a governance gate. High-impact server isolation or service suspension requires human authorization. Low-confidence intrusive action can also be escalated. Thus recommendation authority and execution authority are different objects.

Cells 14 and 15 add an optional bounded LLM supervisory gateway. The notebook runs without it. When enabled, the model receives a compact context and may select only from allowed actions. Deterministic code validates the selection; malformed or unavailable output falls back to deterministic supervision. This follows the general principle that AI components should be bounded by explicit interfaces, independent validation and recoverable failure modes [12; 2; 16].

Illustration omitted in this text-only edition: Cells 11–15 introduce operational consequences, rank defensible actions, separate recommendation from permission, and place any LLM supervision inside a bounded gateway.

OPTIMIZATION WITH JUDGMENT

Optimization answers: *which admissible action scores best under the current representation of objectives?* Judgment asks: *does the current context justify changing the priority ordering among security, continuity, certainty, reversibility and institutional importance?* Governance asks a third question: *even if this is the preferred action, is the machine authorized to execute it?*

The distinction among the three questions is easiest to see by following one recommendation. Suppose an optimizer ranks server isolation first because it produces the largest modeled reduction in threat. That ranking is conditional on the objective weights and the state estimate supplied to it. Judgment examines whether the current context warrants greater emphasis on continuity, confidence or service dependencies than the default ordering provides. The governance gate then checks whether isolation falls within delegated authority. The resulting decision may be to isolate, to choose a reversible alternative, or to escalate. In each case the steps are visible, and a reviewer can disagree with one step without treating the whole system as an opaque agent.

The score is useful as a disciplined comparison, not as a measurement of universal value. Its coefficients express how the demonstration trades defensive benefit against operational burden, reversibility and uncertainty. Changing a coefficient may change the ranking, which is precisely why the values should be reported and tested rather than hidden. Sensitivity analysis can ask whether one action remains preferred across a plausible range of weights or whether the recommendation is fragile. A fragile recommendation is not necessarily wrong; it is a reason to expose the trade-off and perhaps seek human input before acting.

Uncertainty deserves careful treatment because it has two distinct effects. It can make a threat estimate less reliable, and it can make a proposed intervention more dangerous if the system has misunderstood the asset or its dependencies. Penalizing uncertainty may appropriately favor observation or reversible containment in some cases. In other cases, the expected harm of delay may be large enough that uncertainty supports urgent escalation rather than inaction. Therefore, an uncertainty penalty should not be interpreted as a universal instruction to wait. It is one input into a context-sensitive comparison, and the institution should specify what uncertainty means for each action class.

The optional language-model supervisor is best understood as a replaceable adviser behind a narrow interface. It may summarize context or select among pre-approved alternatives, but deterministic validation checks that its answer belongs to the allowed vocabulary and satisfies structural requirements. Failure should lead to a known fallback, not an undefined state or an improvised action. This design also makes comparison possible: the same state can be evaluated with the model enabled and disabled, allowing researchers to determine whether it adds useful contextual discrimination, introduces inconsistency, or merely paraphrases the deterministic recommendation. In none of these cases does its presence change the underlying authorization boundary.

5. Cells 16–20: Execution, Feedback and the Closed Loop

Cell 16 executes only bounded defensive actions. Cell 17 measures institutional availability, critical-service availability and mean health. Cell 18 integrates perception, assessment, judgment, supervision, governance and execution into one SOC decision cycle. Cell 19 provides a clean reset and simulation tick, ensuring that repeated experiments do not inherit stale state. Cell 20 runs a baseline experiment.

This is where the notebook becomes a self-driving system rather than a classifier. Action changes the state; the changed state alters subsequent evidence; subsequent evidence can reorder priorities. Feedback is therefore constitutive, not cosmetic.

Illustration omitted in this text-only edition: Cells 16–20 turn the decision architecture into a closed loop: bounded execution changes the environment, feedback measures institutional condition, and the next decision begins from the revised state.

The control logic can be represented as

$$x_{t+1} = F(x_t, a_t, \varepsilon_{t+1}), \quad y_t = H(x_t, \nu_t),$$

where x_t is the true institutional state, y_t the noisy perceived state, a_t the executed action and ε, ν shocks and observation noise. The institution never acts on x_t directly; it acts on a representation of it. This is why confidence and feedback matter.

Execution should be distinguished from recommendation because a plan that appears sensible before action can have unforeseen effects once applied. A defensive measure may reduce exposure but also remove a service, shift load to another node, or make later observations harder to interpret. The executor therefore needs a defined action interface that checks the target, action type, current state and permission before changing anything. If a precondition is no longer true, the system should stop and reassess rather than executing a stale recommendation. This is the operational equivalent of checking that the world still resembles the one for which the decision was made.

The outcome measures in the notebook are intentionally broader than threat reduction. Institutional availability asks whether the system as a whole remains usable; critical-service availability makes it possible to see whether essential functions are disproportionately affected; mean health offers a coarse summary of condition. No one measure is sufficient. A high mean health can conceal the failure of one critical service, while a temporary availability decline may be acceptable if it prevents a larger loss. Reporting several measures forces the reader to inspect those distinctions instead of treating a single aggregate as proof of success.

Feedback also creates a discipline for revisiting earlier assumptions. After action, the system should ask whether the expected threat reduction occurred, whether disruption was greater than anticipated, whether new telemetry changed confidence, and whether the event spread to connected assets. A model that records only its recommendation but not the result cannot learn whether its cost estimates or action rules were appropriate. In a research setting, this supports a post-decision audit. In an operational setting, it can identify a need to roll back a reversible measure, escalate to a person, or update the situational picture.

The reset mechanism is equally important. Repeated experiments are meaningful only if the starting conditions are controlled or explicitly varied. A clean reset prevents one policy from inheriting the residual effects of another policy's prior actions. For a paired comparison, both policies should face the same initial network and exogenous shock, while their own decisions are allowed to generate different later states. This preserves fairness without pretending that interventions have no consequences. The resulting closed loop is more informative than repeatedly applying different decisions to a fixed snapshot, because the institution's action is itself part of what creates the next state.

Feedback should be designed with a cadence appropriate to the process being modeled. If observations arrive every tick but actions have delayed effects, immediately treating every fluctuation as proof of success or failure can create oscillation. A service may take time to recover, and a measure intended to reduce propagation may first produce a temporary availability decline. The evaluation should therefore distinguish leading indicators, immediate operational effects and longer-run outcomes. This avoids a simplistic loop in which a short-term dip prompts reversal just before the original intervention would have taken effect. Explicit timing assumptions make the example more useful and help readers see where a real control process would need domain-specific calibration.

6. Cells 21–25: Dilemma, Integrated Experiment, Counterfactual and Command Center

Cell 21 constructs the critical-server dilemma explicitly: substantial evidence on a payments server where aggressive containment has serious operational cost. Cell 22 runs the full integrated experiment after a clean reset. Cell 23 visualizes availability and escalations. Cell 24 compares governed judgment with an aggressive threshold policy. Cell 25 exposes the same judgment and governance functions through an interactive Gradio command center.

The counterfactual is especially important. A rule such as “high threat $\Rightarrow$ isolate” may contain risk faster but impose unnecessary operational damage. The governed policy asks what the asset does, what depends on it, how certain the evidence is, whether the action can be reversed and whether the machine has authority. The notebook therefore evaluates not merely detection accuracy but *decision quality under institutional constraints*.

Illustration omitted in this text-only edition: Cells 21–25 expose the central dilemma, execute the integrated experiment, inspect the audit trail, compare aggressive automation with governed judgment and provide an interactive command center.

THE NOTEBOOK'S MOST IMPORTANT RESULT

A high threat score is not an instruction. It is evidence entering a decision. The same distinction becomes fundamental in finance: a widening spread, deposit outflow, currency depreciation or margin call is not by itself an instruction to intervene. It is evidence that must be interpreted in a network, under uncertainty, with explicit objectives and authority.

The critical-server example is designed to make competing losses concrete. If the server supports payments, isolation may create an immediate continuity problem even if it limits a suspected compromise. If it remains connected, propagation may create a larger and less reversible disruption later. The pedagogical point is not that one response is always correct. It is that the same evidence can justify different actions under different service conditions, confidence levels or authority rules. A system that cannot express these contextual distinctions has not solved the decision problem; it has only automated one threshold.

The integrated experiment should be read as a sequence of decisions rather than as a showcase of a final recommendation. At each tick, the state contains both the incident and the condition of the institution. New observations update the evidence; the assessment estimates severity and spread; the judgment stage compares priorities; governance rules determine whether the recommended action can proceed; and execution modifies the state. Visualizations then help a reader see whether the intervention coincided with a change in availability or escalation. The audit trail matters because two runs may produce similar endpoint outcomes through different decision paths.

Counterfactual comparisons are useful only when their assumptions are explicit. A threshold policy may be intentionally aggressive and can serve as a stress case, but it should not be described as a universal baseline for real SOC practice. Likewise, a governed policy should not be declared superior merely because its rules were designed to protect continuity. Researchers should report which outcomes improved, which worsened and how results vary across scenarios. The proper question is not “which policy wins this one run?” but whether a policy produces a more acceptable distribution of outcomes across varied shocks and measurement error.

An interactive command center can help learners inspect the relationships among inputs, recommendations and permissions, but a graphical interface is not itself evidence of sound governance. Controls should make the current scenario and assumptions visible; outputs should show why an action was ranked and whether it was approved; and a reset should clearly distinguish a fresh run from continued operation. The interface should not invite a user to confuse demonstration controls with real operational authority. In the financial analogy, this caution is even more important: an appealing dashboard can make uncertain indicators look definitive. Its purpose should be to support structured review, not to replace it.

7. A General Model of Optimization with Judgment

Let x_t be the institutional state, y_t the perceived state, $\mathcal{A}_t$ the feasible action set, p_t the current priority ordering and $g(\cdot)$ the governance operator. A conventional optimizer selects

$$a_t^* = \arg \min_{a \in \mathcal{A}_t} C(a \mid y_t, p_t).$$

The richer autonomous problem allows context z_t to alter priorities:

$$p_{t+1} = J(y_t, p_t, z_t),$$

then recomputes the preferred action

$$\tilde{a}_t = \arg \min_{a \in \mathcal{A}_t} C(a \mid y_t, p_{t+1}),$$

and finally applies institutional authority:

$$a_t = g(\tilde{a}_t, \text{rules, limits, mandate}).$$

The state then evolves and the process repeats. This resembles model predictive control in its receding-horizon logic [14], but adds an institutional layer: the hierarchy among legitimate objectives may itself change.

SEQUENCE IS AN OUTPUT, NOT A SACRED INPUT

An approved sequence was rational conditional on an earlier state. Environmental change can make mechanical adherence to that sequence inconsistent with the objective it was intended to serve. Judgment is the controlled process by which the institution decides whether priorities should now take precedence and reorder action.

This formulation also clarifies what judgment is *not*. It is not unconstrained intuition. It operates over measured state, feasible alternatives and explicit consequences. Governance then limits which judgments can become actions.

The notation separates four objects that are often conflated. The state x_t is the underlying condition of the institution, including relationships that may not be observed. The estimate y_t is what the decision process can infer from available signals, with whatever uncertainty and missingness remain. The feasible set $\mathcal{A}_t$ contains actions that can actually be considered at that time, given physical, operational and legal constraints. The priority vector p_t expresses the relative importance assigned to legitimate objectives. Keeping these objects separate prevents a system from treating a measurement as reality, or a recommendation as a feasible and authorized action.

The objective function $C(a \mid y_t, p_t)$ is conditional rather than universal. It can combine expected threat reduction, service disruption, reversibility, delay and uncertainty, but those terms need not be equally important in every setting. Judgment J operates on evidence and context to determine whether the current priority ordering should change. It should not manufacture new objectives or silently alter the action set. If a new objective is proposed, such as protecting a newly identified critical function, the institution should record who approved that change and why. In that sense, judgment is a controlled update to the decision frame, not an exemption from it.

The governance operator g is not another optimizer. It applies rules about mandate, delegation, thresholds and required approvals. It can reject an otherwise attractive recommendation, route it for human review, or permit execution within a limited scope. Separating governance from ranking makes it possible to distinguish a technically preferred option from a permissible action. The distinction matters in public institutions because authority is allocated by law and policy, not inferred from predictive accuracy. A model's confidence that an intervention would be useful does not confer authority to carry it out.

The repeated cycle resembles receding-horizon control because each decision uses an updated estimate and the institution plans again after observing what happened. Its distinctive institutional feature is that priorities and permissions are made explicit parts of the cycle. This creates a practical research agenda: specify how state estimates are formed, how priority changes are triggered, how recommendations are ranked, what governance rules apply, and how outcomes revise future decisions. A formal model does not eliminate ethical or political disagreement. It makes the disagreement concrete enough to inspect, test and assign to the people or institutions responsible for resolving it.

One practical benefit of this formal separation is diagnostic clarity. If two scenarios produce different actions, the researcher can determine whether the cause was a changed state estimate, a different feasible action set, a deliberate priority update or a governance restriction. This is more informative than observing only that an output changed. It also supports sensitivity analysis: hold the state fixed and vary the priority vector, then hold priorities fixed and vary confidence or authority. The resulting comparisons reveal whether the decision is robust or depends on a narrow combination of assumptions.

The model also makes room for non-numeric rules. Some constraints are naturally represented as hard exclusions,

such as an action that is outside delegated authority; others may be soft trade-offs, such as preferring a more reversible option when expected benefits are close. Mixing these two forms carelessly can produce misleading results. A hard legal or safety boundary should not be traded away because a score is high, while a soft preference should not be disguised as an absolute prohibition. The institution must explain which conditions define feasibility and which enter the objective, then test how those choices shape the selected action.

8. Transfer Learning: What Exactly Transfers from Cyber to Finance?

The phrase *transfer learning* is used here in an architectural sense. We are not claiming that a model trained on cyber telemetry can be deployed directly as a bank supervisor. What transfers is the decomposition of the problem and the discipline of decision making.

SOC concept	Financial-stability analogue	Transferable question
Asset criticality	Systemically important bank, CCP, payment system, market maker	What cannot fail without disproportionate consequences?
Dependency graph	Interbank claims, collateral chains, funding links, payment obligations	Where can stress propagate?
Telemetry	Deposits, spreads, haircuts, flows, liquidity, intraday payments	What is the current evidence?
Threat score	Composite stress indicator	How severe is the situation, conditional on confidence?
Propagation pressure	Contagion / fire-sale / run dynamics	Is local stress becoming systemic?
Isolation	Ring-fencing, liquidity restrictions, resolution action, market intervention	What intervention contains risk and what does it disrupt?
Operational cost	Credit contraction, market impairment, payment disruption, fiscal cost	What damage can the defense itself cause?
Reversibility	Temporary facility vs. resolution / closure / capital action	Can the intervention be undone?
Governance gate	Supervisory mandate, central-bank authority, resolution authority	Who is authorized to act?
Feedback	Post-intervention flows, prices, liquidity and confidence	Did the action stabilize or amplify the system?

The transfer is plausible because systemic risk is fundamentally relational. Eisenberg and Noe show how obligations connect financial institutions into a clearing network [8]. Brunnermeier and Pedersen show how market liquidity and funding liquidity can reinforce one another [6]. Adrian and Shin emphasize the endogenous relationship between leverage and financial conditions [1]. These mechanisms make the financial system a dynamic network rather than a collection of independent balance sheets.

Architectural transfer means carrying over a disciplined workflow, not copying variables or code without interpretation. A cyber detector's anomaly score has no direct financial meaning, and a model calibrated on endpoint behavior cannot be presumed to recognize liquidity stress. What can transfer is the separation of observation, inference, consequence estimation, action comparison, authorization and post-action review. Each financial domain must define its own data, causal assumptions, action set and mandate. Researchers should therefore describe the transfer as a design hypothesis and then validate each component against domain-specific evidence.

The mapping in the table is intentionally functional rather than literal. A financial institution is not a server, and a supervisory restriction is not network isolation. Both, however, raise questions about criticality, dependency, side effects, reversibility and authority. The analogy helps a learner ask these questions systematically while also showing where it breaks. For example, market participants can anticipate interventions and change behavior, whereas many synthetic cyber incidents in the notebook are generated independently of the defense. Finance therefore requires a strategic response model if the aim is to study policy consequences rather than simply classify states.

Transfer also depends on the unit of analysis. A cyber SOC may observe assets at a fine time scale and issue actions to individual nodes. A financial-stability authority may receive delayed or aggregated data, share responsibilities with other agencies, and face actions whose legal or market consequences extend across institutions. The state representation must reflect those differences. It may need to distinguish a bank's solvency from its liquidity, intraday payment capacity from end-of-day ratios, and local exposures from common asset holdings. Missingness and latency should be modeled explicitly because they shape what a supervisor can know at decision time.

The most useful transfer exercise is consequently comparative. Begin with the same abstract stages, then instantiate them twice: once for the synthetic SOC and once for a clearly specified financial scenario. For each stage, identify what remains invariant and what must be redesigned. A dependency graph is a shared idea, but its edges have domain-specific meanings; confidence is a shared concern, but its calibration process differs; governance is always required, but the relevant authorities and permitted actions change. Stating both continuity and non-equivalence prevents the analogy from becoming a rhetorical shortcut and turns it into a testable research program.

Architectural transfer means carrying over a disciplined workflow, not copying variables or code without interpretation. A cyber detector's anomaly score has no direct financial meaning, and a model calibrated on endpoint behavior

cannot be presumed to recognize liquidity stress. What can transfer is the separation of observation, inference, consequence estimation, action comparison, authorization and post-action review. Each financial domain must define its own data, causal assumptions, action set and mandate. Researchers should therefore describe the transfer as a design hypothesis and then validate each component against domain-specific evidence.

The mapping in the table is intentionally functional rather than literal. A financial institution is not a server, and a supervisory restriction is not network isolation. Both, however, raise questions about criticality, dependency, side effects, reversibility and authority. The analogy helps a learner ask these questions systematically while also showing where it breaks. For example, market participants can anticipate interventions and change behavior, whereas many synthetic cyber incidents in the notebook are generated independently of the defense. Finance therefore requires a strategic response model if the aim is to study policy consequences rather than simply classify states.

Transfer also depends on the unit of analysis. A cyber SOC may observe assets at a fine time scale and issue actions to individual nodes. A financial-stability authority may receive delayed or aggregated data, share responsibilities with other agencies, and face actions whose legal or market consequences extend across institutions. The state representation must reflect those differences. It may need to distinguish a bank's solvency from its liquidity, intraday payment capacity from end-of-day ratios, and local exposures from common asset holdings. Missingness and latency should be modeled explicitly because they shape what a supervisor can know at decision time.

The most useful transfer exercise is consequently comparative. Begin with the same abstract stages, then instantiate them twice: once for the synthetic SOC and once for a clearly specified financial scenario. For each stage, identify what remains invariant and what must be redesigned. A dependency graph is a shared idea, but its edges have domain-specific meanings; confidence is a shared concern, but its calibration process differs; governance is always required, but the relevant authorities and permitted actions change. Stating both continuity and non-equivalence prevents the analogy from becoming a rhetorical shortcut and turns it into a testable research program.

9. Banking Regulation as a Self-Driving Supervisory System

Consider a supervisory system observing a population of banks. The normal sequence may be: collect regulatory data, calculate ratios, rank institutions, conduct reviews, request remediation and escalate only after thresholds are crossed. A shock can make that sequence obsolete within hours.

Suppose one institution experiences unusual deposit outflows. Initially the correct action may be enhanced monitoring. If outflows accelerate, collateral haircuts rise and interbank counterparties withdraw lines, the same institution can become a propagation node. The supervisory priority ordering changes:

routine compliance → liquidity preservation → contagion containment → resolution readiness.

This is not merely a larger value of the same objective function. It can be a change in which objective dominates.

Diamond and Dybvig's bank-run model demonstrates why individually rational withdrawal behavior can generate unstable aggregate outcomes [7]. Global-games approaches show how strategic uncertainty can shape attacks on institutions or regimes [11]. In such settings, the regulator's problem resembles the SOC: acting too late permits propagation; acting too aggressively can validate panic, impair market discipline or impose unnecessary costs.

SUPERVISORY JUDGMENT

The financial analogue of "do not isolate the payments server automatically" is "do not treat a stress threshold as a self-executing supervisory command." The threshold should trigger a structured judgment process that considers confidence, criticality, network position, intervention cost, reversibility, moral hazard and legal authority.

BCBS 239 is relevant because dynamic supervision depends on timely, accurate and aggregable risk data [5]. Model-risk guidance is relevant because the decision architecture must distinguish model output from management action and preserve independent challenge [16]. The Basel liquidity framework provides another natural layer: liquidity metrics can be inputs to judgment without becoming the whole decision [4].

The supervisory system should also distinguish monitoring from intervention. A change in a liquidity indicator can justify faster data collection, a targeted examination or a request for a contingency plan without automatically implying that restrictions or public support are warranted. This graduated response preserves options while evidence develops. When action is needed, the choice should be proportionate to the likely harm and the quality of the state estimate. Temporary reporting requirements or enhanced monitoring may be easier to reverse than a broad restriction, but they can still impose costs and should have a clear purpose, duration and review point.

An effective supervisory design needs to state how priorities can change and what evidence is relevant to that change. Acceleration of outflows, deterioration in collateral, missed payments, counterparty pullbacks and stress at connected entities may together indicate that a local issue is becoming systemic. Yet a checklist should not become a substitute for contextual analysis. The scale, timing and concentration of exposures matter, and some observations may be correlated or delayed. The system should report the pattern it sees, identify data gaps and indicate what additional observation could distinguish competing explanations.

The architecture also supports a useful distinction between prevention and crisis management. During ordinary periods, the system can emphasize data quality, resilience planning and early remediation. Under stress, priorities may shift toward continuity, containment and preparation for resolution. A well-designed institution does not discard routine objectives; it makes the temporary reprioritization explicit and defines conditions for returning to normal supervision. That transition needs governance because extraordinary measures can have lasting distributional effects even when they are introduced during an emergency.

Finally, self-driving supervision should be evaluated for both speed and legitimacy. Faster triage can help officials direct attention to emerging problems, but an apparently efficient ranking can reproduce data gaps or systematically disadvantage institutions whose business models differ from the training sample. Human review, independent validation and a traceable explanation remain necessary. The objective is not to mechanize supervision wholesale. It is to use computational support to organize evidence and alternatives while retaining responsibility with the people and bodies that possess the relevant mandate.

The supervisory system should also distinguish monitoring from intervention. A change in a liquidity indicator can justify faster data collection, a targeted examination or a request for a contingency plan without automatically implying that restrictions or public support are warranted. This graduated response preserves options while evidence develops. When action is needed, the choice should be proportionate to the likely harm and the quality of the state estimate. Temporary reporting requirements or enhanced monitoring may be easier to reverse than a broad restriction, but they can still impose costs and should have a clear purpose, duration and review point.

An effective supervisory design needs to state how priorities can change and what evidence is relevant to that change.

Acceleration of outflows, deterioration in collateral, missed payments, counterparty pullbacks and stress at connected entities may together indicate that a local issue is becoming systemic. Yet a checklist should not become a substitute for contextual analysis. The scale, timing and concentration of exposures matter, and some observations may be correlated or delayed. The system should report the pattern it sees, identify data gaps and indicate what additional observation could distinguish competing explanations.

The architecture also supports a useful distinction between prevention and crisis management. During ordinary periods, the system can emphasize data quality, resilience planning and early remediation. Under stress, priorities may shift toward continuity, containment and preparation for resolution. A well-designed institution does not discard routine objectives; it makes the temporary reprioritization explicit and defines conditions for returning to normal supervision. That transition needs governance because extraordinary measures can have lasting distributional effects even when they are introduced during an emergency.

Finally, self-driving supervision should be evaluated for both speed and legitimacy. Faster triage can help officials direct attention to emerging problems, but an apparently efficient ranking can reproduce data gaps or systematically disadvantage institutions whose business models differ from the training sample. Human review, independent validation and a traceable explanation remain necessary. The objective is not to mechanize supervision wholesale. It is to use computational support to organize evidence and alternatives while retaining responsibility with the people and bodies that possess the relevant mandate.

10. Speculative Attacks Instead of Cyber Attacks

The notebook becomes especially interesting when the “attacker” is not malicious code but strategic market behavior. A speculative attack can involve investors reallocating portfolios in response to beliefs about reserves, policy credibility, exchange-rate sustainability, liquidity or the behavior of other investors. The system is adversarial in an economic sense: agents react to policy and to one another.

A stylized transfer can be organized as follows. Telemetry becomes exchange-rate pressure, forward premia, reserve loss, cross-currency basis, sovereign spreads, deposit flows and market depth. Perception distinguishes ordinary volatility from coordinated stress. Threat assessment estimates whether pressure is local or propagating. Judgment compares intervention alternatives: communicate, provide liquidity, adjust collateral terms, conduct auctions, alter intervention intensity, coordinate agencies or escalate to extraordinary authority. Governance determines which institution may take which action. Feedback observes whether spreads, flows and market depth stabilize.

Morris and Shin’s work on currency attacks and coordination under incomplete information is particularly useful here because it shows that policy effectiveness depends partly on beliefs about others’ actions [11]. The regulator is therefore not controlling a passive physical system. Its action changes the information environment and can alter strategic behavior.

THE ENDOGENEITY PROBLEM

In cyber simulation, the environment can be scripted. In markets, agents observe the regulator. A liquidity facility, guarantee, intervention or public statement can change beliefs and therefore change the state transition itself. Financial transfer requires an explicit strategic-response layer; otherwise the analogy is incomplete.

This strategic feature changes the meaning of telemetry. A change in an exchange rate or spread may reflect new fundamentals, revised beliefs, temporary order imbalance or an expectation that officials will respond. The same observed movement can therefore have different implications depending on market depth, reserve constraints, policy credibility and the timing of information. A perception layer should combine indicators without assuming that any one series gives direct access to underlying beliefs. It should also preserve alternative explanations so that the decision maker can see where the inference is uncertain.

An intervention can reduce immediate pressure yet create later incentives. A facility may provide liquidity to otherwise viable participants, but if its availability is predictable and poorly bounded, market participants may alter their risk taking. A public guarantee may calm a run in one setting while increasing expectations of future support in another. These are not arguments against intervention; they show why operational benefit, distributional consequences and longer-run incentives belong in the evaluation. The relevant comparison is among feasible policies and their likely response paths, not between intervention and an imagined neutral baseline.

For that reason, a financial experiment should distinguish at least three layers of uncertainty: uncertainty about the current state, uncertainty about how the system evolves, and uncertainty about how strategic actors respond to policy. The first concerns measurement and diagnosis. The second concerns mechanisms such as funding needs or fire-sale feedback. The third concerns beliefs and adaptation. Combining all three into one confidence score would obscure which kind of uncertainty is driving the decision. A structured model can instead report these dimensions separately and specify whether the response is more data collection, a reversible measure, a contingency plan or escalation.

The analogy also raises a communication problem. Announcing an intervention can itself reveal official information or signal a view of institutional strength. Silence can also be interpreted. A regulator’s communication policy should therefore be treated as part of the action set, with attention to timing, audience, consistency and authority. In a synthetic simulator, these effects can be represented with simplified belief updates; the purpose would not be to forecast real investors precisely, but to test whether a policy’s apparent success depends on assuming that agents remain passive. If it does, the result should be described as conditional, not general.

This strategic feature changes the meaning of telemetry. A change in an exchange rate or spread may reflect new fundamentals, revised beliefs, temporary order imbalance or an expectation that officials will respond. The same observed movement can therefore have different implications depending on market depth, reserve constraints, policy credibility and the timing of information. A perception layer should combine indicators without assuming that any one series gives direct access to underlying beliefs. It should also preserve alternative explanations so that the decision maker can see where the inference is uncertain.

An intervention can reduce immediate pressure yet create later incentives. A facility may provide liquidity to otherwise viable participants, but if its availability is predictable and poorly bounded, market participants may alter their risk taking. A public guarantee may calm a run in one setting while increasing expectations of future support in another. These are not arguments against intervention; they show why operational benefit, distributional consequences and longer-run incentives belong in the evaluation. The relevant comparison is among feasible policies

and their likely response paths, not between intervention and an imagined neutral baseline.

For that reason, a financial experiment should distinguish at least three layers of uncertainty: uncertainty about the current state, uncertainty about how the system evolves, and uncertainty about how strategic actors respond to policy. The first concerns measurement and diagnosis. The second concerns mechanisms such as funding needs or fire-sale feedback. The third concerns beliefs and adaptation. Combining all three into one confidence score would obscure which kind of uncertainty is driving the decision. A structured model can instead report these dimensions separately and specify whether the response is more data collection, a reversible measure, a contingency plan or escalation.

The analogy also raises a communication problem. Announcing an intervention can itself reveal official information or signal a view of institutional strength. Silence can also be interpreted. A regulator's communication policy should therefore be treated as part of the action set, with attention to timing, audience, consistency and authority. In a synthetic simulator, these effects can be represented with simplified belief updates; the purpose would not be to forecast real investors precisely, but to test whether a policy's apparent success depends on assuming that agents remain passive. If it does, the result should be described as conditional, not general.

11. Other Financial Applications of the Same Architecture

Liquidity crisis management

A bank treasury allocates cash and collateral across payments, margin calls, secured funding and customer withdrawals. The efficient morning plan can become dangerous after a large margin shock. Priorities must be reordered while hard constraints remain intact. This is a direct analogue of the SOC's changing action sequence.

Central clearing counterparties

A CCP monitors member exposures, margin, default resources and market liquidity. A default-management action that protects the CCP can itself affect market liquidity. Judgment must balance containment, auction design, concentration and systemic consequences.

Sovereign and market stress

A central bank or debt-management office can face simultaneous objectives involving market functioning, inflation credibility, reserve preservation and financial stability. A predetermined intervention sequence may be dominated by new information.

Credit committees and loan workouts

The normal sequence—monitor, covenant review, repricing, collateral request, restructuring, exit—can change after a sectoral shock. A model can optimize expected loss, but judgment decides whether preserving a viable borrower or reducing concentration deserves priority.

Resolution and recovery planning

Resolution is an extreme case of governance-gated action. Options can be modeled ex ante, but execution authority is tightly constrained. Reversibility is low, making confidence, legal mandate and escalation especially important.

These applications share a common structure, but the stakes and time horizons differ. Treasury operations may require decisions within minutes to preserve settlement, while a credit workout may evolve over months and allow repeated negotiation. A CCP may have rule-based procedures triggered by a member default, while sovereign-market stress can involve policy choices whose effects are ambiguous and politically consequential. The architecture should therefore be treated as a template for asking questions, not as a universal workflow with identical thresholds or action timing in every institution.

In liquidity crisis management, a useful state representation would distinguish cash, eligible collateral, encumbrance, expected inflows, payment deadlines and contingent outflows. A model may identify a shortfall under a particular scenario, but judgment assesses whether the scenario is credible, which obligations are critical, and whether a proposed transfer or funding source merely shifts scarcity to another part of the institution. A reversible contingency can be preferable while uncertainty is high, but not if waiting would make contractual or settlement obligations impossible to meet. Governance identifies who can approve funding, collateral substitution or business restrictions.

For central counterparties, the decision context includes both member-level exposure and market-wide liquidity. Default management that maximizes recovery for the CCP in isolation could intensify pressure on surviving members or disrupt trading in related markets. A structured decision process should compare the consequences across participants, test concentration under alternative auction or hedging conditions, and identify which choices are prescribed by rules versus left to authorized discretion. The same principle applies to sovereign and market stress: authorities should distinguish support for orderly functioning from a commitment to defend a particular price or guarantee investor returns.

Credit and resolution settings illustrate how reversibility changes the burden of justification. Restructuring terms may be revisited as borrower information improves, whereas exit, enforcement or resolution can be costly to reverse. A model can estimate expected loss or compare recovery scenarios, yet it cannot decide on its own whether preserving a viable borrower, protecting depositors, maintaining critical functions or minimizing public exposure should dominate. Decision records should make the selected objective and the applicable mandate explicit. Across all these examples, the transferable lesson is to model the institution and its dependencies, compare bounded options, explain priority changes and preserve clear responsibility for execution.

These applications share a common structure, but the stakes and time horizons differ. Treasury operations may require decisions within minutes to preserve settlement, while a credit workout may evolve over months and allow repeated negotiation. A CCP may have rule-based procedures triggered by a member default, while sovereign-market stress can involve policy choices whose effects are ambiguous and politically consequential. The architecture should

therefore be treated as a template for asking questions, not as a universal workflow with identical thresholds or action timing in every institution.

In liquidity crisis management, a useful state representation would distinguish cash, eligible collateral, encumbrance, expected inflows, payment deadlines and contingent outflows. A model may identify a shortfall under a particular scenario, but judgment assesses whether the scenario is credible, which obligations are critical, and whether a proposed transfer or funding source merely shifts scarcity to another part of the institution. A reversible contingency can be preferable while uncertainty is high, but not if waiting would make contractual or settlement obligations impossible to meet. Governance identifies who can approve funding, collateral substitution or business restrictions.

For central counterparties, the decision context includes both member-level exposure and market-wide liquidity. Default management that maximizes recovery for the CCP in isolation could intensify pressure on surviving members or disrupt trading in related markets. A structured decision process should compare the consequences across participants, test concentration under alternative auction or hedging conditions, and identify which choices are prescribed by rules versus left to authorized discretion. The same principle applies to sovereign and market stress: authorities should distinguish support for orderly functioning from a commitment to defend a particular price or guarantee investor returns.

Credit and resolution settings illustrate how reversibility changes the burden of justification. Restructuring terms may be revisited as borrower information improves, whereas exit, enforcement or resolution can be costly to reverse. A model can estimate expected loss or compare recovery scenarios, yet it cannot decide on its own whether preserving a viable borrower, protecting depositors, maintaining critical functions or minimizing public exposure should dominate. Decision records should make the selected objective and the applicable mandate explicit. Across all these examples, the transferable lesson is to model the institution and its dependencies, compare bounded options, explain priority changes and preserve clear responsibility for execution.

12. Research Design: How to Test Whether Judgment Adds Value

A compelling demonstration should not define judgment as successful by construction. The notebook's counterfactual in Cell 24 is a useful beginning: compare the governed policy with a more aggressive threshold rule on the same observations. A stronger financial research design would run many common-shock scenarios under competing policies and allow each policy's actions to generate its own subsequent state.

Useful outcome variables include loss containment, continuity of critical services, false interventions, time to stabilization, fiscal or balance-sheet cost, number of escalations, reversibility, contagion size and tail loss. The experiment should also record model provenance and fallback frequency. If a language model is enabled, one should ask whether it changes decisions selectively at genuinely ambiguous margins or merely restates the deterministic ranking.

A simple evaluation vector is

$$Q(\pi) = (L_{system}, D_{critical}, C_{intervention}, F_{false}, T_{stabilize}, E_{human}),$$

for policy π . There need not be a single scalar winner. The point of judgment is precisely that institutions face legitimate trade-offs. A Pareto comparison may be more informative than one synthetic score.

PROPOSED FINANCIAL EXPERIMENT

Build 25 synthetic banks connected through deposits, interbank claims, securities holdings, collateral and payment obligations. Introduce a funding shock. Allow withdrawals, asset sales and mark-to-market effects to propagate. Compare (i) a fixed threshold policy, (ii) a deterministic multi-objective optimizer, and (iii) optimization with bounded judgment and a governance gate. Measure containment, credit and payment continuity, intervention cost, contagion, false positives and escalation. Then introduce strategic depositor or investor response so policy itself changes the environment.

The experiment should begin with a written protocol that fixes the question, policies, shock families and evaluation criteria before results are inspected. Otherwise, a researcher can unintentionally tune the scenario generator or choose a metric that favors the preferred policy. A policy is a complete decision rule, including its data inputs, thresholds, priority updates, escalation conditions, authorized actions and fallback behavior. Comparing only the ranking formula would omit important differences in how policies handle missing data, human review and failed execution.

Paired scenarios help isolate policy effects. Each policy should begin from an equivalent synthetic network and receive the same exogenous shock path. Once decisions differ, however, the subsequent state should be allowed to diverge according to each policy's actions and the model's transition rules. That is the feature that makes the comparison meaningful: containment, service disruption or asset sales are not merely reported outcomes but mechanisms that influence later conditions. Multiple random seeds and scenario variants can then show whether apparent benefits are robust or depend on one favorable realization.

The outcome vector should include both intended benefits and possible harms. Threat containment and time to stabilization matter, but so do interruption of critical services, false interventions, balance-sheet or fiscal cost, contagion, escalation burden and the number of decisions that required rollback. Researchers should define each measure operationally before running the experiment. For example, "stabilization" needs a time window and a criterion; otherwise a brief improvement can be mistaken for durable recovery. Likewise, false intervention should mean an action taken when the modeled condition did not warrant it under the experiment's stated assumptions, not simply an action the researcher dislikes.

Judgment's contribution can be evaluated through ablation. Compare the full architecture with variants that hold priorities fixed, omit the uncertainty term, disable governance escalation, remove feedback, or replace adaptive judgment with a deterministic threshold. These comparisons reveal which component changes decisions and outcomes. They also help identify cases where judgment adds no value or makes performance worse. A credible result should report those cases. If the advantage appears only when the judgment module receives information unavailable to the baselines, the comparison is not fair; all policies should have access to equivalent observations, with differences restricted to the decision logic being tested.

Finally, qualitative review remains useful alongside quantitative measures. Experts can inspect a sample of decision traces and judge whether the stated rationale matches the available evidence, whether a priority change was proportionate, and whether escalation occurred at an appropriate point. That review should use pre-defined criteria and record disagreement rather than treating expert opinion as infallible ground truth. Together, reproducible simulation, sensitivity analysis, ablation and trace review can establish whether the architecture improves decisions in the modeled environment. They cannot by themselves establish real-world supervisory effectiveness, which requires further validation, legal review and careful study of institutional behavior.

13. Governance: Why Judgment Must Not Become Discretion Without Boundaries

The most important lesson from the SOC architecture is institutional separation. Perception should not silently become execution. Recommendation should not imply permission. A model that ranks interventions should not determine its own mandate.

This matters acutely in financial regulation because actions can redistribute wealth, alter market prices, change competitive positions and create fiscal exposure. The relevant safeguards include explicit action vocabularies, authority matrices, confidence thresholds, independent validation, human escalation for high-impact actions, immutable audit trails, reproducible state snapshots and post-action review. NIST's AI Risk Management Framework emphasizes governance and lifecycle risk management [12]; human-AI interaction research likewise stresses calibrated reliance and visible system status [2].

BOUNDED AUTONOMY

A mature autonomous regulator is not one that acts without humans. It is one that knows which decisions can be automated, which require judgment, which require legal authority, and when uncertainty itself is a reason to escalate.

There is also a Goodhart-style warning: once a supervisory indicator becomes a target or a known trigger, institutions and markets may adapt to it. Static thresholds can therefore degrade as decision rules. Judgment and governance do not eliminate this problem, but they make the adaptation visible and reviewable [9].

Governance should be specified at the level of individual action classes. Observation and routine monitoring may be delegated broadly; temporary controls may require defined limits and an expiry or review condition; actions that interrupt critical services or impose material costs may require human approval; and irreversible or legally consequential measures may be reserved for officials with explicit authority. The particular allocation depends on institutional mandate, but the system should encode it in a form that can be inspected. A generic "human in the loop" label is not enough if no one knows what the person must review, what information they receive, or whether they can meaningfully reject the recommendation.

The record for a consequential decision should preserve the state snapshot, source data or data references, model and policy versions, recommendation, confidence, alternatives considered, approval identity, execution result and subsequent outcome. This allows an independent reviewer to reconstruct not only what the system did but what was knowable at the time. Auditability must not be confused with explainability alone: a plausible narrative is not a substitute for the underlying evidence and decision trace. Conversely, raw logs without an intelligible account of the relevant trade-offs may be too difficult to use. Effective oversight needs both a faithful record and a clear explanation tied to it.

Fallback behavior should be treated as part of governance rather than as a technical afterthought. Data outages, model timeouts, malformed outputs, disagreement among components or a sudden shift outside the tested domain should lead to a pre-defined safe mode. That mode might continue passive monitoring, restrict actions to a low-impact subset or require immediate escalation. The appropriate choice depends on the risk of inaction versus erroneous action. A safe fallback is not necessarily "do nothing"; it is a deliberately selected response whose authority and limitations are known in advance.

Independent validation should test more than average predictive performance. It should examine data quality, robustness to distribution shifts, the effects of correlated errors, action side effects, subgroup impacts, calibration and the behavior of fallback mechanisms. Red-team exercises can probe whether known indicators invite gaming or whether seemingly harmless inputs can trigger disproportionate interventions. Human operators also need training and interface support so that they can recognize uncertainty, challenge a recommendation and understand when the system has changed its priority ordering. Oversight fails if the interface encourages automatic acceptance or if the responsible person lacks time and expertise to review.

These safeguards do not imply that human decisions are always better or unbiased. The governance objective is to allocate responsibility clearly, create challenge and appeal mechanisms, and constrain the consequences of error. For high-impact decisions, institutions should identify who owns the policy, who approves changes, who can halt operation, and who reviews incidents after the fact. A judgment-capable system is institutionally acceptable only when its discretion is bounded, its failures are observable, and its mandate can be revised by an accountable authority rather than by the system itself.

14. Conclusion

The autonomous SOC is a compact model of a much larger idea. The system observes a changing environment, constructs a representation of what is happening, estimates propagation, evaluates feasible actions, prices the cost of intervention, exercises bounded judgment, checks authority, acts and learns from the result. Its most important design choice is the refusal to equate a score with an instruction.

That architecture transfers naturally to finance because financial institutions are dynamic networks in which intervention itself can create consequences. The relevant “attack” may be a deposit run, speculative attack, liquidity spiral, margin cascade or loss of market confidence. In each case, the central problem is not only optimization. It is the controlled reordering of priorities when the environment changes.

The conceptual progression is therefore:

Prediction → Optimization → Judgment → Governance → Execution → Feedback

Optimization remains indispensable: it measures, ranks and preserves feasibility. Judgment does not replace it. Judgment determines when the context makes the previous ordering of objectives inappropriate. Governance prevents that judgment from becoming unbounded discretion. Feedback reveals whether the institution’s action improved or worsened the state.

For banking regulation and systemic risk, this suggests a promising research program: construct self-driving supervisory laboratories in which models do not merely forecast distress but must decide, under explicit mandates, how priorities change as stress propagates. The scientific question is not whether AI can produce a recommendation. It is whether an architecture of *optimization with judgment* can improve institutional decisions while preserving accountability, authority and systemic resilience.

The argument developed across the sections can be summarized as a sequence of responsibilities. First, build an adequate representation of the institution and its dependencies. Second, distinguish observed telemetry from inferred evidence and estimated threat. Third, define feasible actions and make their operational consequences comparable. Fourth, allow context to challenge a previously reasonable ordering of objectives, but require that the reason for any change be made explicit. Fifth, check authority before execution, then observe how the action changes the state. Each responsibility addresses a different failure mode, so collapsing them into one score would remove precisely the distinctions that make the system reviewable.

This framework does not claim that judgment can be fully automated or reduced to a formula. It makes a narrower claim: the conditions under which priority ordering changes can be made more explicit, recorded and tested. A system may identify that a critical service, low confidence or high propagation potential changes the trade-off; a human or authorized institution may still need to decide what that trade-off means. The model’s contribution is to organize the evidence, expose alternatives and show consequences under stated assumptions. It should not be credited with values or authority that belong to the institution.

The synthetic SOC is valuable because it allows students and researchers to vary those assumptions safely. One can change dependency structure, scenario severity, action reversibility, confidence or governance rules and inspect how recommendations respond. The same laboratory logic can support financial applications, provided the financial state and behavior are modeled in their own terms and the limits of the analogy are documented. A financial simulator should include strategic response, institutional mandates, data latency and intervention side effects rather than assuming that agents remain passive or that one authority controls every action.

The next research step is therefore empirical and institutional, not merely computational. Researchers should validate the data and transition assumptions, compare policies across diverse scenarios, examine who bears intervention costs, and test whether decision makers can understand and challenge the system under realistic time pressure. Strong evidence would show not only that the architecture improves selected outcomes, but also when it fails, what safeguards limit those failures, and whether responsibility remains clear. That is the standard implied by optimization with judgment: disciplined comparison without false certainty, adaptive priorities without unbounded discretion, and autonomy that remains answerable to the institution it serves.

References

- [1] T. Adrian and H. S. Shin, “Liquidity and Leverage,” *Journal of Financial Intermediation*, vol. 19, no. 3, pp. 418–437, 2010.
- [2] S. Amershi et al., “Guidelines for Human-AI Interaction,” *Proceedings of CHI*, 2019.
- [3] K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press, 2010.
- [4] Basel Committee on Banking Supervision, *Basel III: The Liquidity Coverage Ratio and Liquidity Risk Monitoring Tools*. Bank for International Settlements, 2013.
- [5] Basel Committee on Banking Supervision, *Principles for Effective Risk Data Aggregation and Risk Reporting (BCBS 239)*. Bank for International Settlements, 2013.
- [6] M. K. Brunnermeier and L. H. Pedersen, “Market Liquidity and Funding Liquidity,” *Review of Financial Studies*, vol. 22, no. 6, pp. 2201–2238, 2009.
- [7] D. W. Diamond and P. H. Dybvig, “Bank Runs, Deposit Insurance, and Liquidity,” *Journal of Political Economy*, vol. 91, no. 3, pp. 401–419, 1983.
- [8] L. Eisenberg and T. H. Noe, “Systemic Risk in Financial Systems,” *Management Science*, vol. 47, no. 2, pp. 236–249, 2001.
- [9] C. A. E. Goodhart, “Problems of Monetary Management: The U.K. Experience,” in *Papers in Monetary Economics*, Reserve Bank of Australia, 1975.
- [10] J. G. March and H. A. Simon, *Organizations*. Wiley, 1958.
- [11] S. Morris and H. S. Shin, “Unique Equilibrium in a Model of Self-Fulfilling Currency Attacks,” *American Economic Review*, vol. 88, no. 3, pp. 587–597, 1998.
- [12] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, 2023.
- [13] I. Rahwan et al., “Machine Behaviour,” *Nature*, vol. 568, pp. 477–486, 2019.
- [14] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., 2017.
- [15] H. A. Simon, *Administrative Behavior*. Macmillan, 1947.
- [16] Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, *Supervisory Guidance on Model Risk Management (SR 11-7 / OCC 2011-12)*, 2011.